

Exponential Equations and Logarithmic Equations

Name
Date
Course

Exponent Rules Learned in Pre-Algebra

I) $x^a \cdot x^b = x^{a+b}$ Example: $x^2 \cdot x^5 = x^{2+5} = x^7$

II) $\frac{x^a}{x^b} = x^{a-b}$, $x \neq 0$ Example: $\frac{x^9}{x^3} = x^{9-3} = x^6$

New Exponent Rules

III) $(xy)^a = x^a \cdot y^a$ Example: $(2m)^4 = 2^4 \cdot m^4$

IV) $\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$, $y \neq 0$ Example: $\left(\frac{n}{3}\right)^7 = \frac{n^7}{3^7}$

V) $(x^a)^b = x^{ab}$ Example: $(x^{10})^3 = x^{10 \cdot 3} = x^{30}$

Strange New Exponent Rules

VI) $x^1 = x$ Example: $3^1 = 3$

VII) $x^0 = 1$, $x \neq 0$ Example: $5^0 = 1$

VIII) $x^{-a} = \frac{1}{x^a}$, $x \neq 0$ Example: $x^{-2} = \frac{1}{x^2}$

IX) $x^{\frac{1}{a}} = \sqrt[a]{x}$ Example: $x^{\frac{1}{3}} = \sqrt[3]{x}$

X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$ or $\sqrt[a]{x^b}$ Example: $x^{\frac{5}{7}} = (\sqrt[7]{x})^5$ or $\sqrt[7]{x^5}$

Exponential Equations that Require Exponent Rules X and VIII'

Solve. Write the Roman numeral of each exponent rule that you use.

Rule X) $x^{\frac{b}{a}} = (\sqrt[a]{x})^b$
or $\sqrt[a]{x^b}$

Rule VIII') $\frac{1}{x^{-a}} = x^a$

① $\sqrt[3]{36^5} = \frac{1}{6^{-x}}$

X $36^{\frac{5}{3}} = \frac{1}{6^{-x}}$

VIII' $36^{\frac{5}{3}} = 6^x$

$(6^2)^{\frac{5}{3}} = 6^x$

V $6^{2(\frac{5}{3})} = 6^x$

$6^{\frac{10}{3}} = 6^x$

$\frac{10}{3} = x$

② $\frac{1}{49^{-x}} = \sqrt[8]{7^3}$

X $\frac{1}{49^{-x}} = 7^{\frac{3}{8}}$

VIII' $49^x = 7^{\frac{3}{8}}$

$(7^2)^x = 7^{\frac{3}{8}}$

$7^{2x} = 7^{\frac{3}{8}}$

$2x = \frac{3}{8}$

$x = \frac{3}{8} \div 2$

$x = \frac{3}{16}$

$$\textcircled{3} \frac{1}{2^{-2+x}} = (\sqrt[3]{16})^x$$

$$\text{X} \quad \frac{1}{2^{-2+x}} = 16^{\frac{x}{3}}$$

$$\text{VIII}' \quad 2^{-1(-2+x)} = 16^{\frac{x}{3}}$$

$$2^{2-x} = 16^{\frac{x}{3}}$$

$$2^{2-x} = (2^4)^{\frac{x}{3}}$$

$$\text{V} \quad 2^{2-x} = 2^{4(\frac{x}{3})}$$

$$2^{2-x} = 2^{\frac{4x}{3}}$$

$$2-x = \frac{4x}{3}$$

$$\frac{2}{1} - \frac{x}{1} = \frac{4x}{3}$$

$$\frac{6}{3} - \frac{3x}{3} = \frac{4x}{3}$$

$$\frac{6-3x}{3} = \frac{4x}{3}$$

$$6-3x = 4x$$

$$6 = 7x$$

$$\boxed{\frac{6}{7} = x}$$

$$\textcircled{4} (\sqrt[4]{2})^x = \frac{1}{4^{-3x-2}}$$

$$\text{X} \quad 2^{\frac{x}{4}} = \frac{1}{4^{-3x-2}}$$

$$\text{VIII}' \quad 2^{\frac{x}{4}} = 4^{-1(-3x-2)}$$

$$2^{\frac{x}{4}} = 4^{3x+2}$$

$$2^{\frac{x}{4}} = (2^2)^{3x+2}$$

$$\text{V} \quad 2^{x/4} = 2^{2(3x+2)}$$

$$2^{x/4} = 2^{6x+4}$$

$$x/4 = 6x+4$$

$$\frac{x}{4} = \frac{6x}{1} + \frac{4}{1}$$

$$\frac{x}{4} = \frac{24x}{4} + \frac{16}{4}$$

$$\frac{x}{4} = \frac{24x+16}{4}$$

$$x = 24x+16$$

$$-23x = 16$$

$$\boxed{x = -\frac{16}{23}}$$

Converting from Logarithmic Equations to Exponential Equations

Review

Example I) Solve: $7^x = 20$ Example II) Solve: $3^x = 9$

$$x = \log_7 20$$

$$x \approx 1.5395$$

Calculator required.

Method A

$$3^x = 9$$

$$3^x = 3^2$$

$$x = 2$$

Method B

$$3^x = 9$$

$$x = \log_3 9$$

$$x = 2$$

Calculator not required, but can be used.

When the value of a logarithm is simple (e.g. $x=2$ in Method B above), it can be found without a calculator, but you must create a logarithmic equation and convert it to an exponential equation.

Find the value of each logarithm without a calculator.

⑤ $\log_8 1$

$$x = \log_8 1$$

$$8^x = 1$$

$$8^x = 8^0$$

$$x = 0$$

$$\text{So, } \boxed{\log_8 1 = 0}$$

⑥ $\log_2 16$

$$x = \log_2 16$$

$$2^x = 16$$

$$2^x = 2^4$$

$$x = 4$$

$$\text{So, } \boxed{\log_2 16 = 4}$$

⑦ $\log_5 125$

$$x = \log_5 125$$

$$5^x = 125$$

$$5^x = 5^3$$

$$x = 3$$

$$\text{So, } \boxed{\log_5 125 = 3}$$

⑧ $\log_8 64$

$$x = \log_8 64$$

$$8^x = 64$$

$$8^x = 8^2$$

$$x = 2$$

$$\text{So, } \boxed{\log_8 64 = 2}$$

⑨ $\log_2 8$

$$x = \log_2 8$$

$$2^x = 8$$

$$2^x = 2^3$$

$$x = 3$$

$$\text{So, } \boxed{\log_2 8 = 3}$$

⑩ $\log_4 64$

$$x = \log_4 64$$

$$4^x = 64$$

$$4^x = 4^3$$

$$x = 3$$

$$\text{So, } \boxed{\log_4 64 = 3}$$

Logarithmic Equation.
Exponential Equation.
VII

$$\textcircled{11} \log_3 81$$

$$x = \log_3 81$$

$$3^x = 81$$

$$3^x = 3^4$$

$$x = 4$$

$$\text{So, } \boxed{\log_3 81 = 4}.$$

$$\textcircled{12} \log_5 25$$

$$x = \log_5 25$$

$$5^x = 25$$

$$5^x = 5^2$$

$$x = 2$$

$$\text{So, } \boxed{\log_5 25 = 2}.$$

Logarithm Property I

Logarithmic equations and exponential equations are equivalent. In other words, $x = \log_b C$ if and only if $C = b^x$. We learned in geometry class that "if and only if" means the converse of the statement is true also. Note: Traditionally, neither "C" nor "b" is allowed to be negative when placed in a logarithmic expression.

Determine if the following logarithms can be found without a calculator. If so, calculate the values. If you use an exponent rule, write its Roman numeral.

$$\textcircled{13} \log_2 32$$

$$x = \log_2 32$$

Logarithmic Equation.

$$2^x = 32$$

Exponential Equation.

$$2^x = 2^5$$

$$x = 5$$

$$\text{So, } \boxed{\log_2 32 = 5}.$$

No calculator required.

$$\textcircled{15} \log_{10} 1000$$

$$x = \log_{10} 1000$$

$$10^x = 1000$$

$$10^x = 10^3$$

$$x = 3$$

$$\text{So, } \boxed{\log_{10} 1000 = 3}.$$

No calculator required.

$$\textcircled{14} \log_5 7$$

$$x = \log_5 7$$

$$5^x = 7$$

Calculator required.

$$\textcircled{16} \log_9 15$$

$$x = \log_9 15$$

$$9^x = 15$$

Calculator required.

$$(17) \log_3 30$$

$$x = \log_3 30$$

$$3^x = 30$$

Calculator required.

$$(19) \log_4 5$$

$$x = \log_4 5$$

$$4^x = 5$$

Calculator required.

$$(18) \log_8 \sqrt{2}$$

$$x = \log_8 \sqrt{2}$$

$$8^x = \sqrt{2}$$

$$8^x = 2^{\frac{1}{2}}$$

IX

$$(2^3)^x = 2^{\frac{1}{2}}$$

V

$$2^{3x} = 2^{\frac{1}{2}}$$

$$3x = \frac{1}{2}$$

$$x = \frac{1}{6}$$

$$\text{So, } \log_8 \sqrt{2} = \frac{1}{6}$$

No calculator required.

$$(20) \log_3 9$$

$$x = \log_3 9$$

$$3^x = 9$$

$$3^x = 3^2$$

$$x = 2$$

$$\text{So, } \log_3 9 = 2$$

No calculator required.

$$(21) \log_7 1$$

$$x = \log_7 1$$

$$7^x = 1$$

VII

$$7^x = 7^0$$

$$x = 0$$

$$\text{So, } \log_7 1 = 0$$

No calculator required.

$$(23) \log_6 \left(\frac{1}{36}\right)$$

$$x = \log_6 \left(\frac{1}{36}\right)$$

$$6^x = \frac{1}{36}$$

$$6^x = \frac{1}{6^2}$$

VIII

$$6^x = 6^{-2}$$

$$x = -2$$

$$\text{So, } \log_6 \left(\frac{1}{36}\right) = -2$$

No calculator required.

$$(22) \log_9 80$$

$$x = \log_9 80$$

$$9^x = 80$$

Calculator required.

$$(24) \log_2 22$$

$$x = \log_2 22$$

$$2^x = 22$$

Calculator required.

More Properties of Logarithms

Find the value of each logarithm without a calculator.

(25) $\log_3 3^8$

Logarithm Property II

$$\log_b b^x = x$$

$$x = \log_3 3^8$$

$$3^x = 3^8$$

$$x = 8$$

So, $\log_3 3^8 = 8$.

(26) $\log_6 6^4$

$$\boxed{4}$$

(27) $\log_2 2^5$

$$\boxed{5}$$

(28) $\log_4 4^2$

$$\boxed{2}$$

(29) $\log_8 8$

$$x = \log_8 8$$

$$8^x = 8$$

$$8^x = 8^1$$

$$x = 1$$

So, $\log_8 8 = 1$.

Logarithm Property III

$$\log_b b = 1$$

(30) $\log_{16} 16$

$$\boxed{1}$$

(31) $\log_5 5$

$$\boxed{1}$$

(32) $\log_{32} 32$

$$\boxed{1}$$

(33) $2^{\log_2 5}$

$$x = \log_2 5$$

$$2^x = 5$$

$$2^{\log_2 5} = 5$$

Logarithm Property IV

$$b^{\log_b x} = x$$

(34) $9^{\log_9 4}$

$$\boxed{4}$$

(35) $6^{\log_6 3}$

$$\boxed{3}$$

(36) $11^{\log_{11} 8}$

$$\boxed{8}$$

Logarithmic Equations

Solve without a calculator. If you use an exponent rule, write its Roman numeral.

$$(37) \log_4 X = 2$$

$$4^2 = X$$

$$\boxed{16 = X}$$

VI

$$(38) \log_x 9 = 1$$

$$x^1 = 9$$

$$\boxed{X = 9}$$

$$(39) \log_4 16 = X$$

$$4^X = 16$$

$$4^X = 4^2$$

$$\boxed{X = 2}$$

$$(40) 3 = \log_2 X$$

$$2^3 = X$$

$$\boxed{8 = X}$$

$$(41) \log_x 16 = 4$$

$$x^4 = 16$$

$$\sqrt[4]{x^4} = \pm \sqrt[4]{16}$$

$$X = \pm 2$$

$$\boxed{X = 2} \text{ or } X \neq -2$$

V

$$(42) X = \log_{27} 81$$

$$27^X = 81$$

$$(3^3)^X = 81$$

$$3^{3X} = 81$$

$$3^{3X} = 3^4$$

$$3X = 4$$

$$\boxed{X = \frac{4}{3}}$$

$$(43) \log_5 X = 3$$

$$5^3 = X$$

$$\boxed{125 = X}$$

$$(44) 2 = \log_x 9$$

$$x^2 = 9$$

$$\sqrt{x^2} = \pm \sqrt{9} \quad \text{VII}$$

$$X = \pm 3$$

$$\boxed{X = 3} \text{ or } X \neq -3$$

$$(45) \log_4 1 = X$$

$$4^X = 1$$

$$4^X = 4^0$$

$$\boxed{X = 0}$$

$$(46) 1 = \log_3 X$$

$$3^1 = X$$

$$\boxed{3 = X}$$

VI

$$(47) 3 = \log_x 27$$

$$x^3 = 27$$

$$\sqrt[3]{x^3} = \sqrt[3]{27}$$

$$\boxed{X = 3}$$

V

$$(48) X = \log_{16} 64$$

$$16^X = 64$$

$$(4^2)^X = 4^3$$

$$4^{2X} = 4^3$$

$$2X = 3$$

$$\boxed{X = \frac{3}{2}}$$

More Complicated Logarithmic Equations

$$(49) \log_3 X = -\frac{1}{2}$$

$$3^{-\frac{1}{2}} = X$$

$$\frac{1}{3^{\frac{1}{2}}} = X$$

$$\frac{1}{\sqrt{3}} = X$$

$$\boxed{\frac{\sqrt{3}}{3} = X}$$

$$(50) X = \log_2 \sqrt{2}$$

$$2^X = \sqrt{2}$$

$$2^X = 2^{\frac{1}{2}}$$

$$\boxed{X = \frac{1}{2}}$$

$$(51) \log_x \left(\frac{4}{9} \right) = 2$$

$$x^2 = \frac{4}{9}$$

$$\sqrt{x^2} = \pm \sqrt{\frac{4}{9}}$$

$$X = \pm \frac{\sqrt{4}}{\sqrt{9}}$$

$$X = \pm \frac{2}{3}$$

$$\boxed{X = \frac{2}{3}} \text{ or } X \neq -\frac{2}{3}$$

$$(52) \log_3 \left(\frac{1}{9} \right) = X$$

$$3^X = \frac{1}{9}$$

$$3^X = \frac{1}{3^2}$$

$$3^X = 3^{-2}$$

$$\boxed{X = -2}$$

$$(53) \frac{X}{3} = \log_5 \sqrt[3]{5^4}$$

$$5^{\frac{X}{3}} = \sqrt[3]{5^4}$$

$$5^{\frac{X}{3}} = 5^{\frac{4}{3}}$$

$$\frac{X}{3} = \frac{4}{3}$$

$$\boxed{X = 4}$$

$$(54) \log_7 7 = x^2$$

$$7^{x^2} = 7$$

$$7^{x^2} = 7^1$$

$$x^2 = 1$$

$$\sqrt{x^2} = \pm \sqrt{1}$$

$$X = \pm 1$$

$$\boxed{X = 1} \text{ or } \boxed{X = -1}$$